PRICE RECEIPT

GROUND TRUTH

Prices from the ground, paid in USDC

MLH Solana Hackathon ····· Solana devnet

VERIFIED

THE PROBLEM

> **Price data from informal markets is hard to find.**

> **Traders, NGOs, and food importers guess.**

> **The people who see the prices earn nothing for sharing them.**



THE SOLUTION

Photograph a price. AI checks it. You get paid.

CAPTURE

An in-app camera

VERIFY

A vision model and checks

REWARD

0.05 USDC per verified photo, with no SOL needed

HOW IT WORKS

1 CAPTURE

A photo from the in-app camera, with a one-time code valid for 120 seconds.

2 EXTRACT

A vision model reads item, price, currency, and unit as structured data.

3 VERIFY

Checks reject spoofs, duplicates, and unclear photos.

4 REWARD

0.05 mock USDC arrives in the contributor's wallet. The treasury pays the fee.

5 PUBLISH

Verified prices join a public feed and a live map.

6 ACCESS

Buyers and AI agents query the data. The paid query is planned.

IN REVIEW

LIVE DEMO

Then switch to the app.

5 scenes · about 90 seconds

THE CHECKS

LIVE TODAY

- ✓ Live capture with a one-time code that expires in 120 seconds
- ✓ Duplicate detection with a perceptual hash, compared over seven days
- ✓ Screen and print detection by the vision model
- ✓ Confidence review: below 0.7, no payout

PLANNED

- ☐ Geofence
- ☐ Price outlier check
- ☐ Rate limit
- ☐ Paid query at 0.01 USDC

PROOF IT WORKS

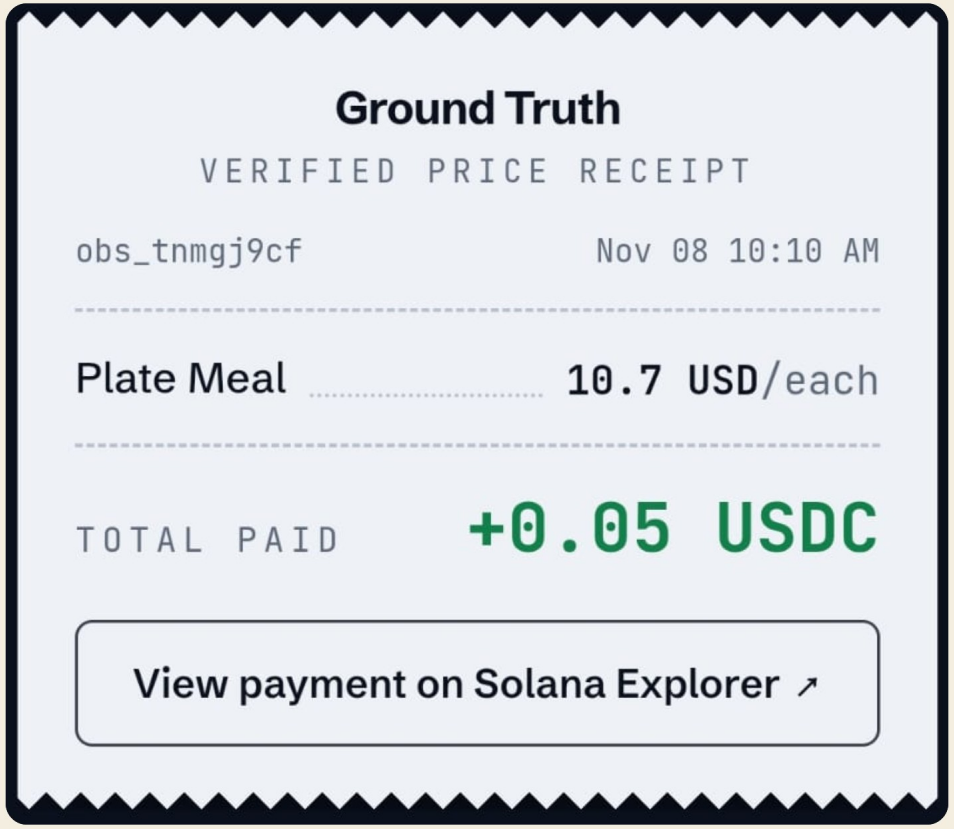
6 of 6 test photos classified correctly

2 of 2 screen photos caught as spoofs

[N] 10 verified photos · 2.5 USDC paid · 2 attempts blocked

Every payout links to a devnet explorer page with its memo.

https://github.com/obaseyii/ground-truth/tree/main



WHY SOLANA

SPONSORED FEES

The contributor holds no SOL. The treasury paid a measured fee of 11,030 lamports, about 0.000011 SOL.

SPEED

About four seconds from photo to verified payout in our tests.

MEMO

A memo on every payout ties the payment to the photo hash.

USDC SETTLEMENT

A 5-cent reward makes sense only when fees are this small.

DEVNET TRANSACTION

FEE PAYER

Treasury

TRANSFER

0.05 mock USDC

MEMO

gt:[observation-id]:[photo-hash]

THE DATA

A public feed of verified prices, each tied to a photo hash and a payout signature

A live map and a price explorer on the project website

PLANNED

A paid query at 0.01 USDC for businesses and AI agents

[Screenshot: website map with real records]

[Screenshot: price explorer with real records]

WHERE WE FIT

PRIOR ART

HIVEMAPPER

Pays contributors for street imagery on Solana.

WHAT WE ADD

GROUND TRUTH

Prices, paid on Solana with sponsored fees, tied to photo hashes, with spoof checks shown live.

PRIOR ART

ORATRUTH

A Bittensor project. Rewards miners for OCR-verified price photos.

KNOWN LIMITS AND NEXT STEPS

KNOWN LIMITS

The app runs on a laptop with a local database.

The token is a mock USDC on devnet and has no value.

Receipts of any date are accepted.

The test set holds six photos.

Item names vary with the model's output.

NEXT STEPS

- [ ] Geofence, price outlier, and rate limit checks
- [ ] A receipt date check
- [ ] Paid queries with x402
- [ ] Name normalization
- [ ] A hosted backend with a real database
- [ ] Solana Attestation Service records for each observation

TEAM AND LINKS

VERIFIED

TEAM

Emmanuel

Backend, vision extraction, Solana payouts, capture app

Mohamed Benabdellah

Project website, video editing

LINKS

WEBSITE

ground-truth-website.vercel.app

CODE

github.com/obaseyii/ground-truth

LIVE APP

nov-mere-invalid-written.trycloudflare.com

Online while the laptop runs

DEMO VIDEO

https://youtu.be/kpIH-8sP08I

PRICES FROM THE GROUND, PAID IN USDC.